

Predicting and Disrupting the MDM2–p53 Protein–Protein Interaction Using AlphaFold2

A computational structural biology project

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Abstract

The binding between p53 and MDM2 is one of the most studied protein–protein interactions in cancer biology, because p53 protects cells from becoming cancerous and MDM2 is the protein that keeps it switched off. In this project I used AlphaFold2, run through the ColabFold platform, to predict MDM2 and p53 first on their own and then as a complex. The predicted complex matched the known crystal structure well, so I then made a single point mutation in p53 (W23A) at one of the key interface residues and predicted the complex again. The interface confidence score (ipTM) fell from 0.690 to 0.400, a 42% drop, which suggests that changing just one residue is enough to break the whole interaction. This agrees with what experiments have already shown about this residue, and it gives a simple example of how AlphaFold can be used not just to predict a complex but to study and target a protein–protein interaction.

1. Introduction

p53 is often called "the guardian of the genome." It is a tumor suppressor: when a cell is damaged or stressed, p53 can stop the cell cycle or even trigger cell death, which stops a damaged cell from turning into cancer. MDM2 is the protein that keeps p53 under control. It binds to p53, blocks its activity, and also tags it for destruction. The problem is that when there is too much MDM2, p53 gets shut down completely, and the cell loses one of its main defenses. This is why blocking the MDM2–p53 interaction has become an important idea in cancer drug development.

The structure behind this interaction is well known. A short piece of p53, the transactivation domain (TAD), folds into a small helix that slots into a hydrophobic pocket on the N-terminal part of MDM2. Three p53 residues do most of the work, and together they are usually called the "hydrophobic triad": Phe19, Trp23, and Leu26. These three side chains reach deep into the MDM2 pocket and hold the two proteins together. Trp23 is the most important of the three — it is the "anchor" residue, because its large indole ring goes the deepest into the pocket.

My aims in this project were:

1. To predict the 3D structures of MDM2 and p53, both alone and together, using AlphaFold2.
2. To score and compare these predictions using AlphaFold's own confidence values.
3. To find which residues form the interaction interface.
4. To make a point mutation at a key residue and check whether the interaction is lost.

This also connects directly to the topic of "targeting protein–protein interactions," which is the research area of this course.

2. Materials and Methods

2.1 Computational environment

I did all of the work in Google Colaboratory (Colab), which is a free online platform for running Python notebooks. I used the Colab Pro version with an L4 GPU and a High-RAM runtime, because AlphaFold simply takes too long without a GPU. To make sure nothing got lost when a session timed out, I connected the notebook to Google Drive and saved every result, figure, and file there as I went.

The actual structure prediction was done with ColabFold (version 1.6.1). ColabFold is a tool that runs AlphaFold2 and AlphaFold2-Multimer (v3) in an easy way, and it speeds up the slowest part — the homology search — by using a fast method called MMseqs2 instead of the original one. After trying to install AlphaFold inside my own notebook and running into version problems, I decided to run the predictions in the official ColabFold notebook, since that is the most stable option, and then do all the analysis afterwards in my own notebook.

2.2 Protein sequences

Rather than cutting short pieces out of the crystal structure, I took the full, real sequences from the UniProt database, so that the predictions would start from biologically correct input:

- p53 (human): UniProt P04637 (393 amino acids)
- MDM2 (human): UniProt Q00987 (491 amino acids) Predicting the entire proteins would have been slow, and the large unstructured regions tend to make the result noisy.

Since the binding happens only in a small part of each protein, I used just the interacting domains:

- p53 transactivation domain: residues 1–73 (73 aa)
- MDM2 N-terminal domain: residues 1–125 (125 aa)

This way I kept the region that actually matters for the interaction and left out the disordered parts that would only add noise.

2.3 Structure predictions

I ran three separate predictions:

1. MDM2 alone (monomer)
2. p53 alone (monomer)
3. MDM2 + p53 together (complex / multimer)

For the complex, I joined the two sequences with a colon (:), which is how ColabFold knows to treat them as two separate chains. Each run produced five ranked models, and I used the top one (rank_001) for the analysis. I kept the default settings (3 recycles, 5 models) so that the runs would be easy to reproduce.

2.4 Mutation

To test how important the anchor residue really is, I changed Trp23 in p53 to alanine (W23A). This swaps the big indole ring of tryptophan for the tiny methyl group of alanine. Only this one residue was changed — the rest of the sequence stayed exactly the same, still 73 amino acids long — so that any difference in the result could be linked to this single change and nothing else. I then predicted the mutant complex (MDM2 + p53-W23A) in exactly the same way as the wild-type.

2.5 Analysis

To judge the predictions, I looked at the confidence scores that AlphaFold reports:

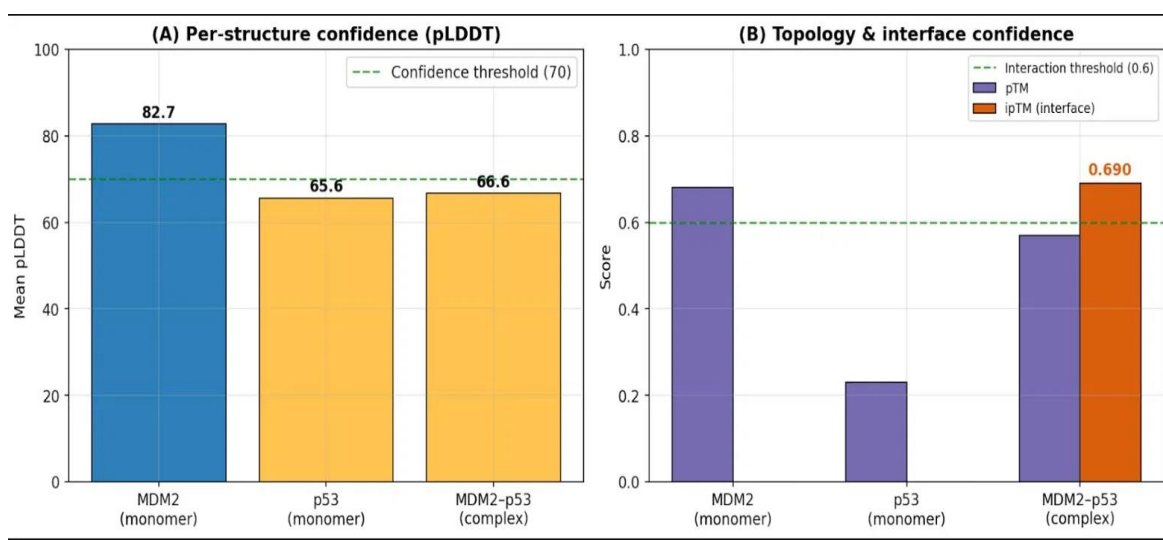
- pLDDT (0–100): how sure the model is about the local structure. Higher is better, and anything above 70 is usually treated as reliable.
- pTM (0–1): confidence about the overall shape (topology) of the model.

- ipTM (0–1): confidence about how the two chains sit relative to each other. This is the key number for an interaction, and a value above 0.6 usually means the complex is reliable.
- PAE (Predicted Aligned Error): the expected error in the distance between pairs of residues. When the error between the two chains (the off-diagonal blocks of the PAE map) is low, the interaction is more reliable.

For the interface itself, I wrote a short script using BioPython that looks for every residue where an atom of one chain comes within 5 Å of an atom of the other chain, and then counts how many atomic contacts each residue makes. This is how I found the interface residues and measured which ones contribute the most.

3. Results

3.1 AlphaFold predicts MDM2 well, p53 poorly, and a confident complex



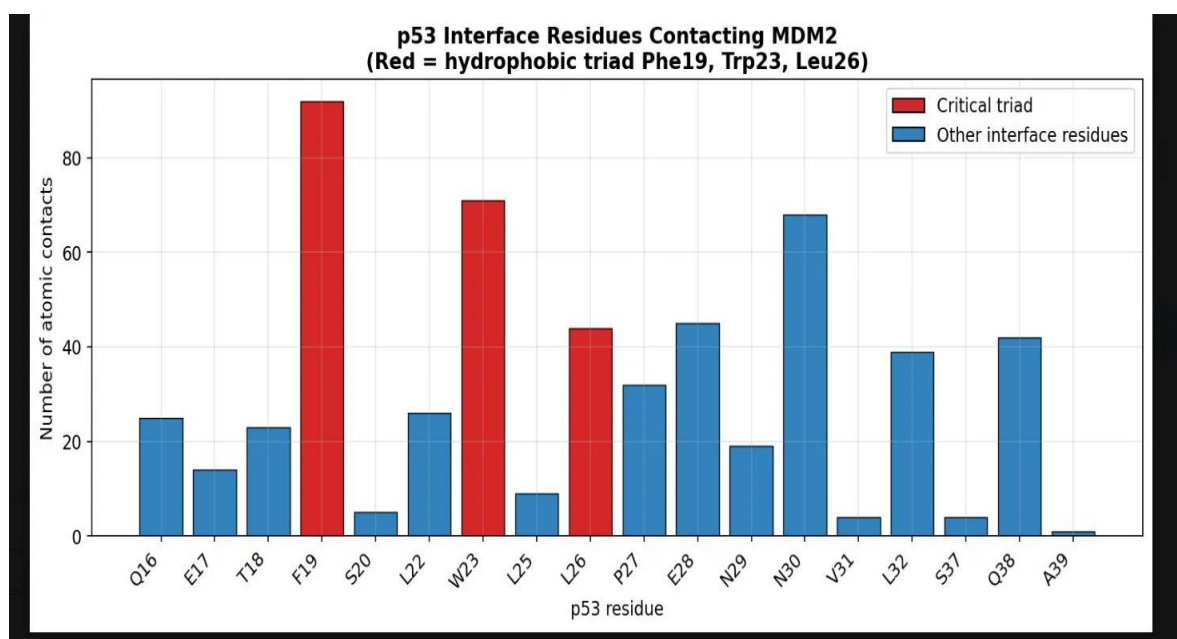
(Figure 1) shows the confidence scores of the three predictions.

MDM2 alone was predicted with high confidence (mean pLDDT = 82.7, pTM = 0.680). This makes sense because the MDM2 N-terminal domain is a stable, well-folded domain. p53 alone had a low score (mean pLDDT = 65.6, pTM = 0.230). This is not a mistake — it is actually an important biological result. The p53 transactivation domain is **intrinsically disordered**,

which means it does not have a fixed shape on its own. It only becomes ordered when it binds to a partner. AlphaFold correctly shows this low confidence.

For the complex, the most important score is **ipTM = 0.690**, which is above the 0.6 threshold. This means AlphaFold predicts that MDM2 and p53 really do interact. This value became my baseline for the mutation experiment.

3.2 The predicted interface matches the known crystal structure



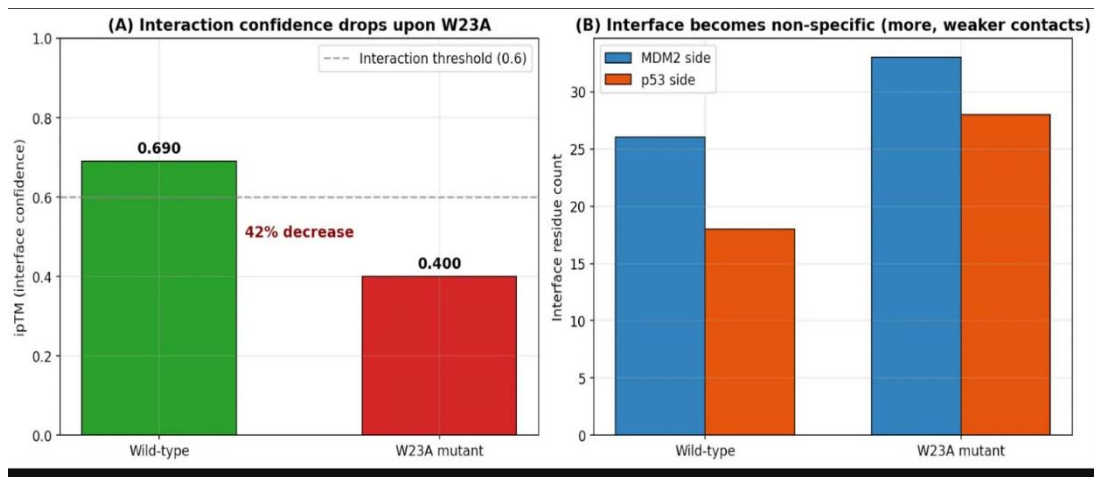
(Figure 2) shows, for each p53 residue, how many atomic contacts it makes with MDM2. The three residues of the hydrophobic triad are marked in red.

All three triad residues are in the interface and make many contacts:

- **Phe19:** 92 contacts (the most)
- **Trp23:** 71 contacts
- **Leu26:** 44 contacts

This is a strong validation. The crystal structure (PDB 1YCR) is known to use exactly these three residues, and AlphaFold predicted the same thing without being given the structure. This means the AlphaFold model is biologically correct and can be trusted as the basis for the mutation experiment.

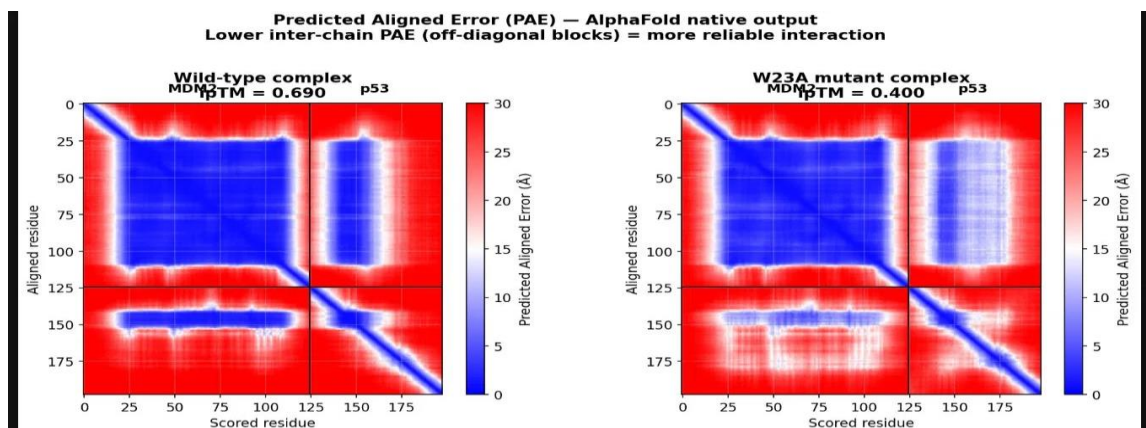
3.3 The W23A mutation disrupts the interaction



(Figure 3A) compares the ipTM of the wild-type and the mutant complex. After the W23A mutation, ipTM dropped from **0.690 to 0.400**, a **42% decrease**. The score went from above the 0.6 threshold (reliable interaction) to below it (uncertain interaction). This is the main result of the project: a single point mutation at the anchor residue is enough to break the interaction.

(Figure 3B) shows an interesting and at first surprising result. The number of interface residues did **not** go down — it actually went up (from 44 to 61 total). This is not a contradiction; it is more evidence that the interaction is broken. In the wild-type, the big Trp23 ring fits into the MDM2 pocket like a key in a lock, giving a small but **specific and strong** binding. When Trp23 becomes the tiny Ala, this lock is lost. p53 can no longer sit correctly in the pocket, so it floats around and touches MDM2 in many **random, non-specific, and weak** places. So there are more contacts, but they do not form a real binding. The quality went down even though the quantity went up — and this is exactly why the ipTM dropped.

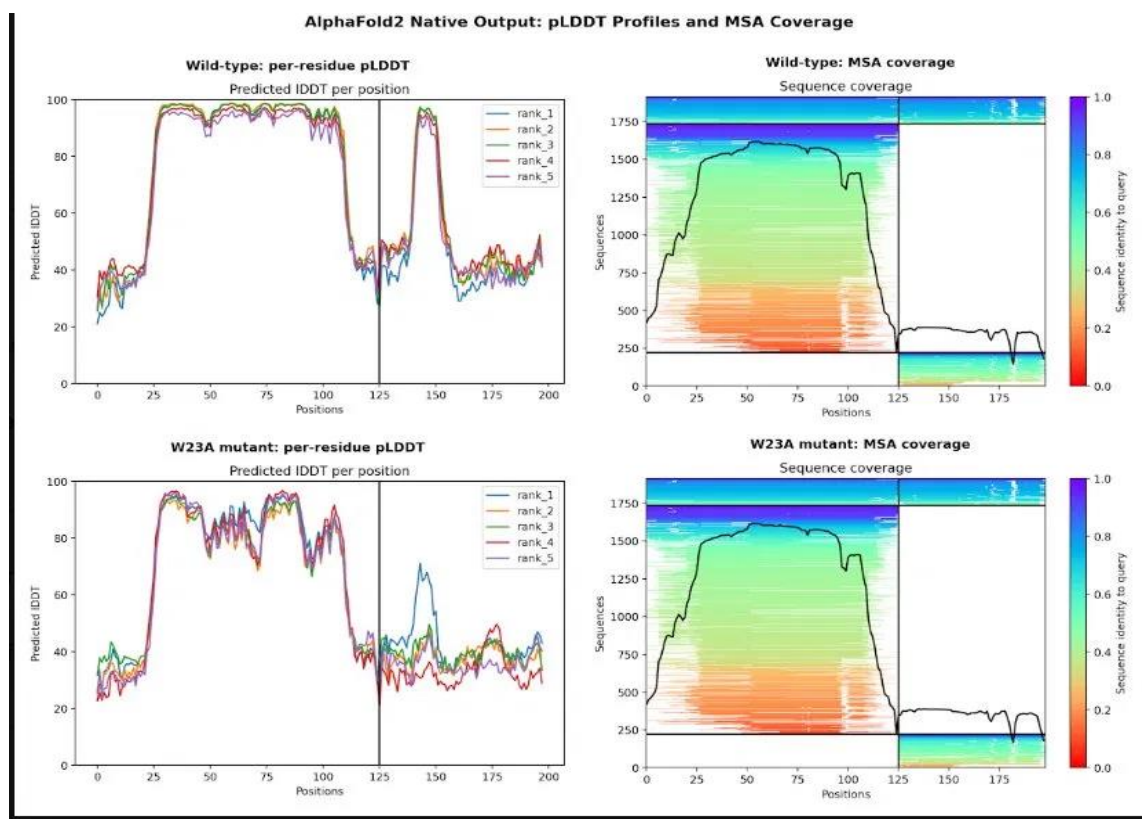
3.4 PAE confirms the loss of interaction (AlphaFold native output)



(Figure 4) shows the Predicted Aligned Error (PAE) maps, which come directly from AlphaFold. The map is divided into four blocks by the chain boundary. The two diagonal blocks (top-left = MDM2, bottom-right = p53) show how well each chain is predicted by itself. The two off-diagonal blocks show the confidence **between** the chains — this is what matters for the interaction.

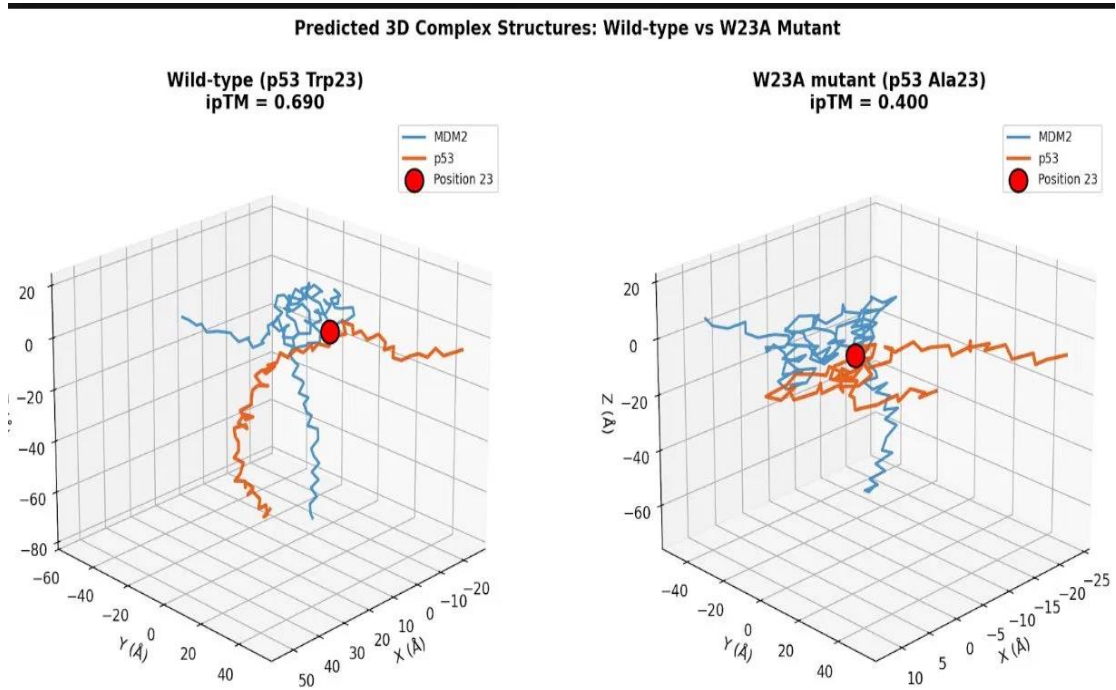
In the wild-type (left), the off-diagonal blocks have clear blue regions, meaning low error between the chains, so the interaction is reliable. In the mutant (right), these off-diagonal regions become much lighter and more red, meaning higher error and a less reliable interaction. This agrees with the drop in ipTM.

3.5 AlphaFold confidence profiles and sequence alignment



(Figure 5) shows AlphaFold's own native output: the per-residue pLDDT curves (left) and the MSA (multiple sequence alignment) coverage (right), for both the wild-type and the mutant. The pLDDT curves show that the MDM2 part (first ~125 residues) is predicted with high confidence in both cases, while the p53 part (residues ~126–198) has lower confidence, again because it is naturally flexible. The MSA coverage plots show that AlphaFold found many similar sequences for MDM2 but fewer for the short p53 part, which is normal for a small disordered region.

3.6 3D structural comparison



(Figure 6) shows the predicted 3D structures of the two complexes side by side, with MDM2 in blue, p53 in orange, and position 23 marked as a red sphere. In the wild-type (left), p53 wraps neatly against the MDM2 domain near the anchor residue. In the mutant (right), the p53 chain looks more loose and disordered around the contact region, which fits with the lower confidence scores.

4. Discussion

In the end, the experiment came down to one amino acid. For the wild-type complex, AlphaFold gave an ipTM of 0.690, and the residues it placed at the interface were the same ones the crystal structure uses — Phe19, Trp23, and Leu26. I did not tell the model anything about the known structure, so the fact that it found the right residues on its own made me trust the prediction.

The mutation result was the part I did not fully expect. After changing only Trp23 to Ala, the ipTM fell to 0.400 — a 42% drop — while everything else in the sequence stayed the same. What surprised me was that the interface did not get smaller; it actually got bigger (44 to 61 residues). At first this looked like a contradiction, but I think it is the opposite. In the wild-type, the large Trp23 ring fits into the MDM2 pocket like a key, so the binding is small but very

specific. Take that key away, and p53 has nothing to lock onto, so it just leans against MDM2 in many loose, random spots. More contacts, but none of them mean anything. That is probably why the confidence went down even though the number of contacts went up — and it is also why ipTM tells you more about a real interaction than just counting contacts does.

This fits with what is already known: Trp23 is described as the anchor residue, and removing it is known to weaken p53 binding a lot. It also lines up with how real cancer drugs work. The nutlins, for example, sit in the same MDM2 pocket and push p53 out — they attack exactly this hotspot. So even though my experiment is small and fully computational, it follows the same logic that real drug design uses.

Of course there are limits to what I did. AlphaFold gives one fixed structure, not the actual movement of binding, and ipTM is only a confidence score — it is not a real measurement of binding strength. I also used only the interacting domains instead of the full proteins, so some context is left out. If I continued this project, I would try other mutations such as F19A or L26A, and compare the predictions with real experimental binding data to see how well they agree.

5. Conclusion

In this project I used AlphaFold2 through ColabFold on Google Colab to predict MDM2 and p53 on their own and as a complex, checked the predicted interface against the known crystal structure, and then broke the interaction with a single mutation (W23A). The ipTM dropped from 0.690 to 0.400, which points to Trp23 being the key residue holding the two proteins together. For me the main takeaway is that AlphaFold is not only useful for predicting what a complex looks like — it can also be used to ask what happens when you change it, which is exactly the kind of question that matters when trying to target protein–protein interactions in cancer.

Figure Legends

Figure 1. Prediction confidence scores. (A) Mean pLDDT for MDM2 alone, p53 alone, and the complex; dashed line = reliability threshold (70). (B) pTM and ipTM scores; ipTM = 0.690 for the complex (above the 0.6 interaction threshold).

Figure 2. Atomic contacts of each p53 interface residue with MDM2 in the predicted wild-type complex. The hydrophobic triad (Phe19, Trp23, Leu26) is shown in red. All three triad residues are present in the interface, matching the crystal structure.

Figure 3. Wild-type versus W23A mutant. (A) ipTM drops from 0.690 to 0.400 (42% decrease) after the mutation. (B) Interface residue counts; the interface becomes larger but non-specific in the mutant.

Figure 4. Predicted Aligned Error (PAE) maps from AlphaFold for the wild-type (left) and mutant (right) complexes. The off-diagonal blocks (between chains) become less confident (more red) in the mutant.

Figure 5. AlphaFold native output: per-residue pLDDT profiles (left) and MSA sequence coverage (right) for the wild-type (top) and mutant (bottom) complexes.

Figure 6. Predicted 3D structures of the wild-type (left) and W23A mutant (right) complexes. MDM2 = blue, p53 = orange, position 23 = red sphere.

References

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